Guards, Squads and Incentives: Keeping Large Teams of Language-Model Agents Effective and Safe

Chuntao Dan

Claude Opus 5.5*
Anthropic

Abstract

Teams of language-model agents trust bad information, pool less of what they know as they grow, and, when trained with the wrong incentives, stop cooperating. In experiments with teams of up to 64 Claude Haiku 5.5 agents and reinforcement learning (RL) of a 3B open model, each analysed under a plan frozen before its runs, we find five things. (1) *Verification works as an assigned role.* A quarter of the team assigned as guards, who post, confirm and contest clues and never solve, recovered 0.72 of the damage done by an equal number of liars, with no ground truth, against 0.34 for a track record computed from the true answers; guards beat the unprotected team at every size from 8 to 64 agents and also against mistaken teammates, colluding liars and Claude agents prompted to deceive. (2) *Splitting the team makes it scale.* Guards can contest only the clues they hold, so their advantage shrank as teams grew; squads of eight, each with its own guards, made it grow instead, from 0.26 at 16 agents to 0.34 at 64. (3) *Larger teams are more gullible.* The share of answers built on a false clue rose from 0.79 at 8 agents to 0.91 at 32, because a posted lie settles a clue, not because lies spread further. (4) *Copies of one model do not divide labour on their own.* Offered the guard role without an assignment, no agent took it; RL with a shared team reward did not make it emerge; and the reward decided whether trained copies kept cooperating. (5) *Move information; do not add work.* Mechanisms that pass on or certify information at no cost in turns helped, while those that add work to every agent's turn or demand obedience backfired. We release the environments, the frozen plans and code that regenerates every number and figure.

1 Introduction

Multi-agent deployments of language models are growing, from orchestrators with parallel workers [10] to large groups working on open-ended problems, and their failures are well documented. On hidden-profile tasks, where the answer needs facts only individual members hold, groups of agents reach about 30% accuracy where one agent with every fact reaches 81% [13], and Anthropic's red team reports 17–36% for most models [2]. Agents conform [27], adopt a single false testimony readily [28], free-ride in public goods games when they are reasoning models [9], and become opportunistic when trained with RL in social dilemmas [21]. The red team names the fixes as open problems: mechanism design, reputation and environments that shape cooperation [2].

We test fixes, and how they behave as teams grow. For each failure we build a small, fully specified environment, freeze an analysis plan before any run, add one mechanism at a time, and pair each mechanism (an *affordance*: an action, an observation line, a role) with a prompt-only *instruction* control (§3). Our five findings, strongest first:

*An AI model made by Anthropic, working as a research and coding agent in Claude Code; see the author contributions.

Preprint.

1. **Verification works as an assigned role** (§4.1): guards recover more of liars' damage than an oracle track record, with no ground truth, at every size from 8 to 64 agents and against every kind of faulty teammate we tested.
2. **Splitting a large team makes verification scale** (§4.2): the guards' advantage shrinks with size because they hold only about two in five clues; squads of eight with their own guards make it grow with size instead.
3. **Larger teams are more gullible, for a traceable reason** (§4.3): exact loss decompositions of replayed runs show agents building on a posted lie more often; lies do not spread further.
4. **Copies of one model do not divide labour on their own, and the reward decides whether they cooperate** (§4.4): no agent volunteers for an unassigned role, team-reward RL does not create one, and individual reward teaches guessing instead of sharing.
5. **Move information; do not add work** (§4.5): a shared board, a coordinator that passes clues on and a public track record helped; a request board, an instruction to share first, private checks and orders to follow a coordinator did not.

2 Related work

Information pooling in agent teams. The hidden-profile paradigm [24] has been ported to LLM groups [2, 13]. Diagnoses stress early convergence on shared evidence [13], averaging of expert and non-expert views [18] and loss of issue-critical facts in deliberation [25]; an LLM facilitator raised how much human groups shared without improving their decisions [1]. Adaptive communication control [3] and structured deliberation [22] can beat unstructured discussion. Pruning messages and agents saves tokens at comparable accuracy [26, 30], and across 260 configurations multi-agent architectures range from large gains to large losses against a single agent [12]; in our tasks a message costs task time, not just tokens.

Deception, trust and fault tolerance. Defences learn credibility scores from past contributions [6], rank neighbours [7] or rectify misleading content [16]. We compare a public, harness-computed signal with private verification at matched cost, and test oracle-free rules borrowed from social insects: quorums [23] and guards [5]. Honest-signalling theory explains when a signal can be trusted, namely when it is costly to fake [29]; our results suggest that who sees the verification, and whose job it is, matter too.

Training agent teams. Multi-agent RL for LLM collaboration includes MAPoRL [19], MAGRPO [15] and AT-GRPO [31]; counterfactual credit addresses free-riding [8, 14]. RL-trained LLM agents become opportunistic in social dilemmas [21] and most societies fail to sustain a commons [20]. Grading that never penalises wrong answers rewards guessing [11]; we observe the multi-agent analogue, a producer-scrounger outcome [4].

3 Methods

The clue board. N agents share a board of $2N + 2$ open subtasks, each worth 1, 2 or 3 points and open for 30 turns. A subtask hides an integer $0 \leq x < 30$ fixed by two clues "$x \bmod p = r$" dealt to different agents, so no agent can solve it alone. Each turn an agent opens a subtask, answers it (a wrong answer costs 0.5), shares a clue, or waits. Shared clues go to a *blackboard* every agent reads; in the channel comparisons they go instead to direct messages, nowhere, or to a coordinator agent that collects them and suggests subtasks. Sharing costs the sharer a turn and earns it nothing directly, so the board poses the hidden-profile problem with a price on communication.

Faulty teammates. For the defences (§4.1–§4.2) each clue is dealt to two agents, so a lie about a clue can be contradicted by its other holder, and a quarter of the agents are *liars*: scripted agents that, every other turn, post a false residue for a subtask they are not working on. We also test *mistaken* teammates, honest agents that misread 30% of their clues; *colluding* liars, who repost each other's false values and so defeat any rule that trusts a value two agents posted; and *Claude liars*, Claude Haiku 5.5 agents told in a private instruction to make their teammates answer wrongly, which can act only through clue posts. In §4.3 and §4.5 each clue has one holder.

Table 1: Mechanisms tested. Kind: A, an affordance the harness provides; I, a prompt-only instruction; R, an assigned role; S, a change of team structure.

Mechanism	What changes	Kind	Section
Blackboard	shared clues go to a board every agent reads (against no channel)	A	4.5
Follow the coordinator	agents are told the coordinator sees every shared clue and its suggestion is usually best	I	4.5
Coordinator passes clues	with each suggestion, the coordinator passes on every clue it has pooled for it	A	4.5
Request board	a new action posts a request for the clues of the agent's open subtask	A	4.5
Share first	"post every clue you hold for an open subtask before working on your own"	I	4.5
Track record	beside each clue, how many of its sender's earlier clues proved true (needs the truth)	A	4.5
Check action	spend a turn to learn privately whether a posted clue is true	A	4.5
Warning	"a teammate may share false clues; cross-check before relying on one"	I	4.5
Agreement counts	the board counts the agents that posted each value and marks it confirmed or contested	A	4.1
Instruction only	the same advice as one sentence, with no counts shown	I	4.1
Guards	a quarter of the team (as many as the liars) is told to post, confirm and contest clues, never to solve	R	4.1
Volunteer offer	the guard role and its rationale are offered to every agent, with no assignment	I	4.4
Squads	teams of eight on their own boards with their own subtasks, two guards each	S	4.2
Failure flags	every posted value that agreed with a failed answer is flagged on the board	A	4.2
Guard copies	one copy of every clue is also dealt to a guard	A	4.2

Agents. Team members are Claude Haiku 5.5 with low effort. Each turn an agent receives a text observation (the open subtasks, its own clues, the board and a memory of its earlier turns) and replies with one action as JSON. Every reply is cached by its request, so reruns are byte-identical and turn logs can be replayed by tracers at no cost. Rule agents written in numpy, which answer any subtask whose clues they know and otherwise share a clue that a teammate needs for the subtask it has open, give a reference level and a forecast frozen before each expensive run (Appendix C).

Mechanisms. Each arm changes one thing (Table 1). Where we can, we pair an affordance, which the harness provides, with an instruction that gives the same advice in one sentence of the prompt, so that an effect of the affordance is not an effect of telling agents what to do.

Training. We trained Qwen2.5-3B-Instruct on the clue board (four agents, blackboard) in teams of copies sharing one LoRA adapter. A supervised warm start imitated the rule agent's moves (about 14,000 turns in teams of one, two and four). RL then used group-relative policy gradients: each step played 8 boards 4 times each; each agent slot's return was compared with that slot's mean over the four plays, and a quarter of the turns with a nonzero advantage were trained on, with a KL penalty to the warm start. The learning rate was steered to a target per-step policy change (a KL of 0.003, measured on re-scored replies), steps that moved the policy more than four times the target were undone, and runs stopped on persistent invalid replies or drift. Rewards were *individual* (an agent's own points less its penalties), *team* (the team's mean), *credit* (the solver keeps half of a subtask's value and the rest goes to the agents whose shared clues it used) and individual with a training-only price of 3 on a wrong answer. Trained models were scored on 15 held-out boards by welfare, whatever their training reward. Later rounds trained on mixed four- and eight-agent teams and, with team reward, on the board with agreement counts and no role instructions.

Measures. *Welfare* is points per agent-turn. The *pooled share* divides welfare by the welfare of the same team with every clue known to all, so 1 is the information ceiling; with liars present we report the honest agents' pooled share. The *honest loss* is the honest agents' welfare without liars

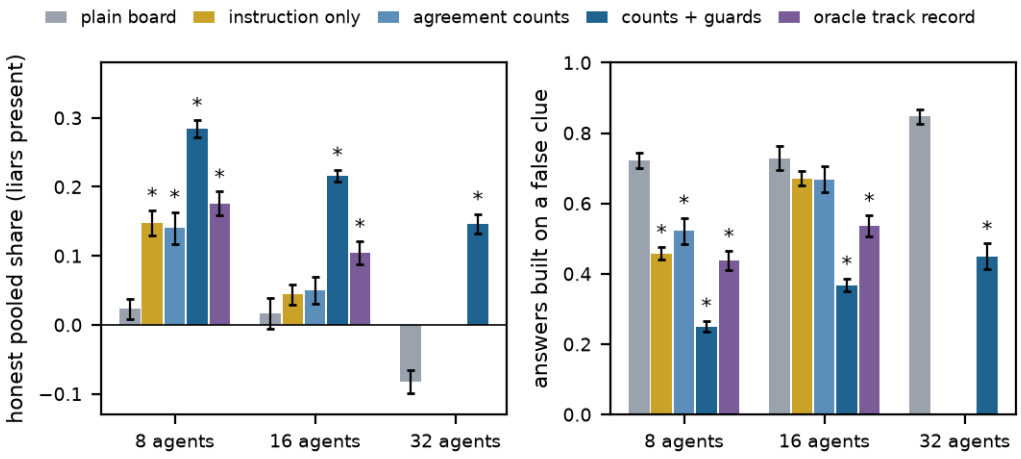


Figure 1: Defences against liars (a quarter of the agents), 8 seeds per bar; at 32 agents only the plain board and guards ran. Left: the honest agents' pooled share. Right: the share of honest answers built on a false clue. Mean ± SEM; * $p < 0.05$ against the plain board, two-sided exact sign-flip test paired on seed, uncorrected.

minus with them, on the same seed and arm, and *loss recovered* is 1− (arm's mean honest loss / baseline's). *Gap closed* is the mean paired difference (arm minus baseline) over the ceiling minus the baseline. *Contamination* is the share of honest answers that used a false clue. *Specialization* is how much more the agents' action mixes differ from one another than in a null that shuffles turns between agent slots.

Design and statistics. Each study's analysis plan (arms, seeds, primary tests and correction family) was frozen, hashed (the first 12 hexadecimal digits of the SHA-256 of its canonical JSON) and committed before its runs (Appendix A); the larger-team stages of §4.1 were declared deviations, each with its own frozen plan. Arms are paired on board seed and compared through their paired differences, as Miller [17] recommends for evaluations, with 5 to 10 seeds per arm (8 for the defences). Tests are paired sign-flip tests over seeds (over training runs when RL designs are compared; exact, or with at least 10,000 random sign flips when the units are too many to enumerate), one-sided where the plan fixed a direction, or exact permutation tests between team sizes, with Holm correction within each plan's primary family. A trend over team size is the mean of each seed's least-squares slope over $\log_2 N$ (per doubling of the team), tested with a two-sided sign-flip test over seeds; trends are descriptive. In the text, brackets give 95% bootstrap intervals over the pairing units. In the figures, points and bars are means and error bars one standard error of the mean (SEM) over seeds or boards (for ratio measures, the bootstrap standard error), and an asterisk marks $p < 0.05$ in a two-sided test against the stated reference, uncorrected. Because several stages were added after earlier results, we treat all findings as exploratory.

4 Results

4.1 Verification works as an assigned role

The setting is the one that defeats a team most easily: one in four agents lies, and the harness does not know the truth. At 8 and 16 agents (8 seeds; plan `58e842277a3e`; $24.89) we compared the plain board, agreement counts, the same advice as an instruction, agreement counts with guards, and an oracle track record of each sender (Figure 1).

Guards worked. They beat the counting board by 0.155 $[0.135, 0.183]$ (Holm p <0.001) and the plain board by 0.26 at 8 agents and 0.20 at 16, recovering 0.72 of the liars' damage against 0.34 for the oracle record. They did it by concentrating checking: guards contested 0.31 false values per lie (counting board 0.21, plain board 0.06), and the share of honest answers built on a lie fell to 0.31 (counting board 0.60, plain board 0.73). **The display did not.** The counting board helped (0.075

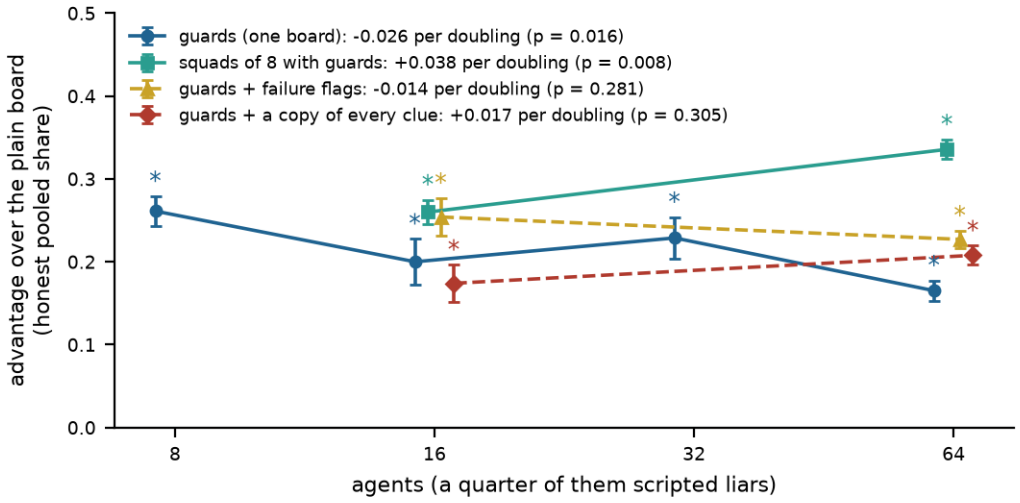


Figure 2: Advantage over the plain board in the honest agents' pooled share against team size, a quarter of the agents scripted liars, 8 seeds per point (the three design changes ran at 16 and 64 agents). Mean ± SEM; * $p < 0.05$, two-sided exact sign-flip test against 0, paired on seed, uncorrected. The legend gives each design's trend: the mean of each seed's slope over $\log_2$ of the team size, per doubling, with its two-sided exact sign-flip p (descriptive). Guards on one board lose ground as teams grow; squads of eight, each with its own guards, gain it.

$[0.048, 0.102]$), but no more than the one-sentence instruction (-0.001), and some honest agents reposted a lie to "confirm" it.

Guards held at every size and against every fault we tested. Guards beat the plain board by $+0.261$, $+0.200$, $+0.229$ and $+0.165$ at 8, 16, 32 and 64 agents, in all eight seeds at each size (one-sided exact $p = 0.004$ each). At 32 agents they also helped against mistaken teammates ($+0.058$ $[0.024, 0.094]$, $p = 0.023$) and against colluding liars ($+0.090$ $[0.054, 0.118]$, $p = 0.008$): a guard contests a wrong value of a clue it holds however many agents posted it. At 16 agents (plan `0c59fa10f2b0`) we replaced the scripted liars with Claude liars. Guards still won ($+0.063$ $[0.016, 0.104]$, Holm $p = 0.039$) and recovered 0.70 of their damage (0.63 for scripted liars). The Claude liars lied less often than scripted ones (an accepted lie in 0.18 of their turns, against 0.5) with the same damage per lie and did not avoid clues a guard held.

The cost. On boards without liars, guards cost 0.09, 0.06 and 0.04 pooled share at 8, 16 and 32 agents: the checking turns are turns not spent solving.

4.2 Splitting a large team makes verification scale

Over 8 to 64 agents the guards' advantage changed by -0.026 per doubling (two-sided $p = 0.016$; descriptive). Replaying every guard board from the response cache showed that the guards themselves did not weaken. At every size they contested 75–79% of the lies on clues they held, with a median delay of about one round, and each such lie led to only 0.19–0.24 wrong answers. The limit is coverage: with two holders per clue, only 37–43% of lies landed on a clue a guard held, and lies on the other clues led to more wrong answers in larger teams (1.49 each at 64 agents) and caused 81–91% of all answers built on a lie.

We then tested three design changes against these guards at 16 and 64 agents (plan and numpy forecast frozen first, `a441381ce078`; $43.80). All three beat them at 64 agents, in all eight seeds (one-sided, Holm $p = 0.012$ each), and on raw welfare too (Figure 2). *Squads* won by $+0.171$ $[0.156, 0.186]$, and their advantage over the unprotected team *grew* with size, from 0.26 at 16 agents to 0.34 at 64, in 8 of 8 seeds ($+0.038$ per doubling, two-sided exact $p = 0.008$; descriptive), while the guards' advantage on one board did not (difference-in-differences against guards $+0.111$ $[0.069, 0.152]$, $p = 0.008$); a 64-agent team of squads kept 0.25 honest pooled share, near the 8-

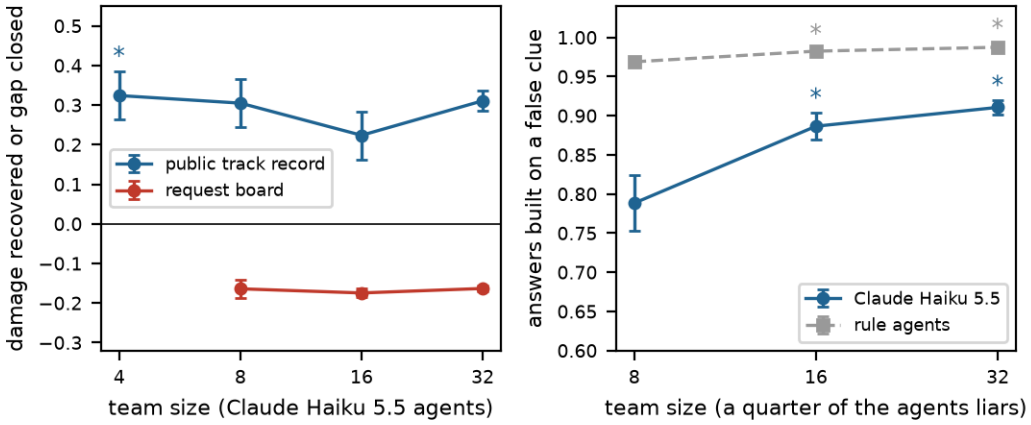


Figure 3: Left: share of the liars' damage recovered by the public track record and share of the pooling gap closed by the request board, against team size (10 seeds at 4 agents, 5 at larger sizes); mean ± bootstrap SE; * $p < 0.05$, two-sided exact sign-flip test against 0 (with five seeds the exact two-sided p cannot fall below 0.0625). Right: share of honest answers built on a false clue, one holder per clue and a quarter of the agents liars, for Claude Haiku 5.5 (5 seeds) and rule agents (20 seeds); mean ± SEM; * $p < 0.05$, exact permutation test against 8 agents.

agent guard board's 0.28, with prompts of 780 tokens instead of 6,463. *Failure flags*, a signal the environment already gives the agent whose answer failed, made public, added $+0.062$ $[0.042, 0.082]$ at 64 agents and $+0.054$ at 16, at no cost on clean boards ($+0.000$), though 0.27 of flags fell on true values. *Guard copies* helped only at 64 agents ($+0.043$ $[0.025, 0.063]$): they cost -0.129 on clean boards at 16, and the guards, now responsible for every clue, contested 0.61 of lies against 0.75. Verification has a capacity limit, and small groups with their own checkers scale where one global board does not. Squads shared no subtasks across squads, so their gain is an upper bound for tasks that split cleanly.

4.3 Larger teams are more gullible

Without a defence, the same share of liars does more damage in a larger team. With one holder per clue and a quarter of the agents liars (5 seeds), the honest agents' loss rose from 0.47 at 8 agents to 0.58 at 16 and 0.54 at 32; with two holders per clue, liars pushed the unprotected team's honest pooled share below zero at 32 and 64 agents (-0.08 and -0.09), with 0.85 of honest answers built on a lie. A numpy rule-agent forecast frozen before the runs predicted the opposite trend.

We replayed the one-holder runs from the response cache with a turn-by-turn tracer (plan `f5c25637acd0`; no API calls) and split the honest loss into additive terms that sum to it exactly. Lies were not more visible in larger teams, and agents almost never passed a liar's clue on (at most 0.25% of liar posts). What grew was how often agents built on a lie: the share of honest answers resting on a false clue rose from 0.79 at 8 agents to 0.91 at 32 (Figure 3, right), such answers were wrong 95–97% of the time, and the loss on subtasks a lie touched carried 0.72 of the total rise. Once a liar had posted a clue, the true clue was never posted afterwards in 89–94% of cases: agents treat a posted clue as settled, whoever posted it. This is the failure guards repair, by contesting the settled clue. The fixes that worked at four agents kept working as teams grew (Figure 3, left): with a quarter of the agents liars, the track record recovered 0.30, 0.22 and 0.31 of the loss at 8, 16 and 32 agents, and the request board cost about the same at every size.

4.4 Copies of one model do not divide labour on their own

Prompting. Offered the guard role with its rationale but no assignment (16 agents), no honest agent acted as a guard on any of 24 boards (0 volunteers): the team did no better than the counting board alone ($+0.009$ $[-0.023, 0.039]$, Holm $p = 0.293$), -0.16 against assigned guards.

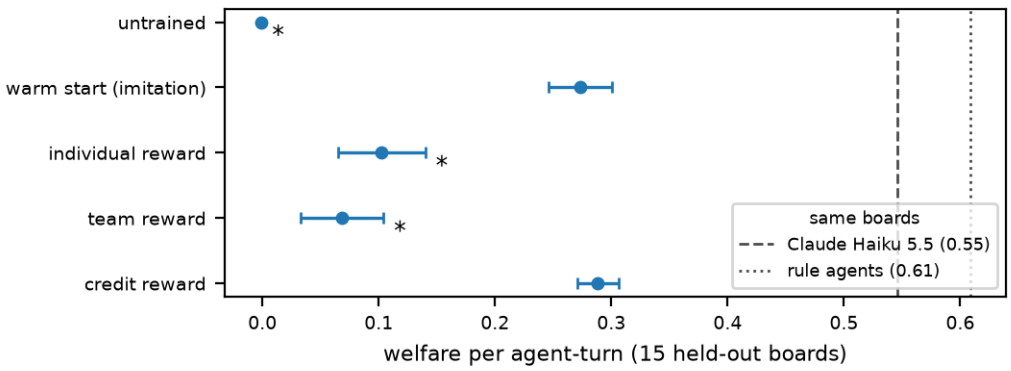


Figure 4: Welfare per agent-turn of trained four-agent teams of Qwen2.5-3B-Instruct on 15 held-out boards: the untrained model, the warm start and one training run per reward, with Claude Haiku 5.5 and rule agents on the same boards for reference. Mean ± SEM over boards; * $p < 0.05$ against the warm start, two-sided exact sign-flip test paired on board, uncorrected.

Training. Imitation did most of the teaching: welfare rose from -0.001 to 0.27 $[0.22, 0.32]$ (Figure 4). The reward decided what RL taught on top. Under individual reward the model stopped sharing (0.92 to 0.15 shares per held clue) and solved by guessing (74% of solves): with a wrong answer costing 0.5, guessing among three candidates for a 3-point subtask earns about 1.25 per turn, more than sharing pays the sharer. Team reward was unstable; credit reward kept sharing and beat individual by 0.19 $[0.09, 0.27]$. The training-only price on wrong answers suppressed guessing in every run (0.1–15 guesses per board against 49–90), but a pre-registered replication with four unpriced and five priced runs found its effect on sharing seed-dependent ($+0.37$ clues per held clue, exact $p = 0.056$, Holm 0.11; the selfish drift in 2 of the four unpriced runs), and we withdrew that claim.

Roles did not emerge. Five runs of 30 steps with team reward on the counting board, with no role instruction (plan `c103584f5aac`), trained stably, but specialization of the agents' action mixes above the shuffled-slot null ($+1.32$, exact $p = 0.0625$) was not significant, and the untrained warm start was already about as specialized ($+3.44$) as the most specialized trained run ($+3.57$). Of the five runs, 2 collapsed to every agent posting and none solving: under team reward a post is never penalized and an answer can be. Training on mixed four- and eight-agent teams (8 runs of 60 steps; plan `8e28b5fdfb1f`) raised welfare at four agents from 0.26 to 0.34 (7 of 8 runs, exact $p = 0.035$, Holm 0.105), but the gain faded with size (0.09 against 0.08 at 32 agents, where Claude Haiku reaches 0.44); trained teams did lose less to liars at 32 agents ($+0.061$ $[0.012, 0.108]$, secondary). Training at small sizes did not teach what large teams need.

A training lesson. Five of those eight runs halted early; four halts were one-step jumps of the logged KL to the warm start while the measured policy change and the team's behaviour stayed flat. The per-token estimate $e^d - d - 1$ (d the reference minus the policy log-probability) is unbounded when the trainer scores a sampled token far below the reference, so a single token moves the mean and, through the same term in the loss, can dominate an update. Capping each token's term and halting only on two consecutive steps removed the false halts.

4.5 Move information; do not add work

Channels and coordinators. A blackboard added 0.39 $[0.37, 0.40]$ welfare per agent-turn over no channel at four agents. A coordinator that routes clues helped little (0.10); telling agents to follow its suggestions made the team worse (-0.13 $[-0.16, -0.10]$), as they opened subtasks they knew nothing about and waited for clues it never passed on, while having it pass on the clues it collected helped (0.12 $[0.07, 0.16]$). A coordinator helps by moving information, not by giving orders.

Communication has a price. The baseline team earned a pooled share of 0.45 at four agents. The request board was used in about a quarter of turns, sharing did not rise, and the gap widened: -0.18 $[-0.21, -0.14]$ at four agents (Holm $p = 0.008$), and about the same at 8, 16 and 32 (-0.16, -0.17,

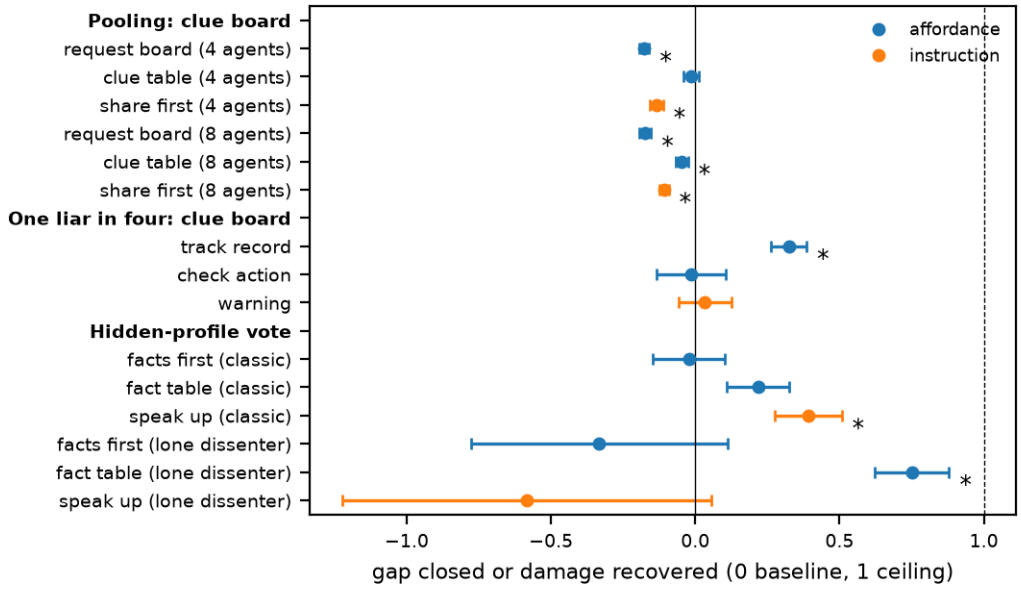


Figure 5: Each fix at four agents (eight where marked): share of the gap to the ceiling closed (pooling, vote) or of the liar's damage recovered (one liar), where 0 is the baseline and 1 the ceiling (dashed). Blue: affordances; orange: prompt-only instructions. Mean ± bootstrap SE over the pairing units (10 seeds on the clue board, 100 episodes in the vote); * $p < 0.05$, two-sided sign-flip test of the paired difference against the baseline (exact over seeds; 10,000 random sign flips over episodes), uncorrected (the text gives the plan's Holm-corrected tests).

-0.16). An instruction to post every held clue first nearly doubled sharing and lowered output (-0.14 $[-0.18, -0.09]$): sharing costs a turn here, and the extra messages spent turns that solving needed (Figure 5).

Public reputation, not private verification. With one liar among four agents, the track record recovered 0.32 $[0.20, 0.44]$ of the honest agents' loss (Holm $p = 0.008$), at no cost without a liar (0.006). The check action caught lies and recovered nothing (-0.01 $[-0.28, 0.19]$); a warning did nothing (0.03). A 2×2 of public or private and free or costly (10 seeds) found that publicity added 0.18 $[0.04, 0.36]$ and being free 0.16 $[0.02, 0.30]$: both matter, about equally. Unlike guards, the record needs the harness to know the truth.

5 Discussion

Where to put the work. The mechanisms that helped made information easier to move or to trust at no cost in turns (a shared board, a public record) or gave checking to agents whose job it was (guards); those that failed added work to every agent's turn (asking, checking, sharing first), demanded obedience, or only displayed what agents could already infer (agreement counts). For harnesses this suggests: put verification in the environment or in a role, not in every agent's turn; and keep each role's span small enough to cover what it must check, by splitting large teams into squads rather than growing one global board.

From assigned to trained roles. Assigned roles worked; offered roles were not taken; and a shared team reward neither created roles nor kept every run productive. Training a division of labour likely needs a reward that pays a contribution when others use it (as credit reward did for sharing) or a role-conditioned objective, and an evaluation against the starting policy, which here was already specialized.

Future directions. Four experiments follow directly. (i) *Train roles rather than reward teams*: a role-conditioned or credit-style reward that pays a guard when its contest prevents a wrong answer,

evaluated against the warm start. (ii) *Squads on coupled tasks*: subtasks whose clues span squads, with bridge agents that carry clues between them, to measure what the squads' gain costs when a task does not split cleanly. (iii) *Stronger adversaries*: Claude liars in 64-agent teams, liars that target the clues no guard holds, and free-text persuasion, against squads and failure flags. (iv) *Correlated errors and real tasks*: teams in which every copy shares a blind spot, guards drawn from a different model, and verification on coding or research tasks where a check costs more than a turn. A better screen would help all four: our rule-agent forecasts missed exactly the behaviours that mattered, such as building on a posted lie, and a simulator that models them could rank designs before any API spend.

Limitations. The tasks are small and synthetic; the Claude teams use one model (Haiku 5.5) and mostly scripted liars; Claude liars could act only through clue posts; correlated errors shared by every copy of a model and compromised agents remain untested. Board arms have 5–10 seeds and RL designs one to eight training runs, which our results show is too few to rank similar designs; RL used a 3B model trained at four and eight agents only. Squads shared no subtasks across squads. The trust fixes have not been tested on a dedicated misinformation benchmark such as ForesightSafety-TIDE [28].

Broader impact. The mechanisms studied are defensive or improve cooperation; the RL results show how easily a plausible incentive teaches agents to abandon cooperation, which matters for anyone training groups of models.

Author contributions

Chuntao Dan directed the project: chose the questions and the order of the studies, made the design decisions recorded in the specifications, ran the GPU training and reviewed every result. Claude Opus 5.5, working as an agent in Claude Code under that direction, wrote the specifications, analysis plans, environments and analysis code, ran the API experiments and replays, analysed the results and drafted the paper. Chuntao Dan takes responsibility for the content.

References

[1] Mohammed Alsobay, David M. Rothschild, Jake M. Hofman, and Daniel G. Goldstein. Bringing everyone to the table: An experimental study of LLM-facilitated group decision making. In *ACM Conference on Computer-Supported Cooperative Work (CSCW)*, 2026. arXiv:2508.08242.

[2] Anthropic Frontier Red Team. Patterns and problems in emerging multiagent systems. https://www.anthropic.com/research/multiagent-systems, 2026.

[3] Abhijith Babu et al. Consilience: Conformally calibrated communication control for hidden-profile multi-agent reasoning, 2026. arXiv:2608.20564.

[4] C. J. Barnard and R. M. Sibly. Producers and scroungers: A general model and its application to captive flocks of house sparrows. *Animal Behaviour*, 29:543–550, 1981.

[5] Michael D. Breed, Ernesto Guzmán-Novoa, and Greg J. Hunt. Defensive behavior of honey bees: Organization, genetics, and comparisons with other bees. *Annual Review of Entomology*, 49:271–298, 2004.

[6] Sana Ebrahimi, Mohsen Dehghankar, and Abolfazl Asudeh. An adversary-resistant multi-agent LLM system via credibility scoring, 2025. arXiv:2505.24239.

[7] Yang Feng and Xudong Pan. SentinelNet: Safeguarding multi-agent collaboration through credit-based dynamic threat detection. In *The ACM Web Conference (WWW)*, 2026. arXiv:2510.16219.

[8] Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. Counterfactual multi-agent policy gradients. In *AAAI Conference on Artificial Intelligence*, 2018.

[9] David Guzman Piedrahita, Yongjin Yang, Mrinmaya Sachan, Giorgia Ramponi, Bernhard Schölkopf, and Zhijing Jin. Corrupted by reasoning: Reasoning language models become free-riders in public goods games. In *Conference on Language Modeling (COLM)*, 2025.

[10] Jeremy Hadfield, Barry Zhang, Kenneth Lien, Florian Scholz, Jeremy Fox, and Daniel Ford. How we built our multi-agent research system. https://www.anthropic.com/engineering/multi-agent-research-system, 2025. Anthropic Engineering, 13 June 2025.

[11] Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. Why language models hallucinate, 2025. arXiv:2509.04664.

[12] Yubin Kim et al. Towards a science of scaling agent systems, 2025. arXiv:2512.08296.

[13] Yuxuan Li, Aoi Naito, and Hirokazu Shirado. Systematic failures in collective reasoning under distributed information in multi-agent LLMs. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2505.11556.

[14] Zhongyi Li et al. Counterfactual credit policy optimization for multi-agent collaboration, 2026. arXiv:2603.21563.

[15] Shuo Liu et al. LLM collaboration with multi-agent reinforcement learning. In *AAAI Conference on Artificial Intelligence*, 2026. arXiv:2508.04652.

[16] Yaoyang Luo et al. Defending LLM-based multi-agent systems against cooperative attacks with sentence-level rectification, 2026. arXiv:2605.28104.

[17] Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations, 2024. arXiv:2411.00640.

[18] Aneesh Pappu, Batu El, Hancheng Cao, Carmelo di Nolfo, Yanchao Sun, Meng Cao, and James Zou. Multi-agent teams hold experts back. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.01011.

[19] Chanwoo Park, Seungju Han, Xingzhi Guo, Asuman Ozdaglar, Kaiqing Zhang, and Joo-Kyung Kim. MAPoRL: Multi-agent post-co-training for collaborative large language models with reinforcement learning. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[20] Giorgio Piatti, Zhijing Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. Cooperate or collapse: Emergence of sustainable cooperation in a society of LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[21] Dereck Piche, Mohammed Muqeeth, Milad Aghajohari, Juan Duque, Michael Noukhovitch, and Aaron Courville. Learning robust social strategies with large language models, 2025. arXiv:2511.19405.

[22] Sunil Prakash. From debate to deliberation: Structured collective reasoning with typed epistemic acts, 2026. arXiv:2603.11781.

[23] Thomas D. Seeley. *Honeybee Democracy*. Princeton University Press, 2010.

[24] Garold Stasser and William Titus. Pooling of unshared information in group decision making: Biased information sampling during discussion. *Journal of Personality and Social Psychology*, 48(6):1467–1478, 1985.

[25] Herun Wan et al. The deliberative illusion: Diagnosing factual attrition and stance homogenization in multi-agent LLM deliberation, 2026. arXiv:2606.03032.

[26] Zhexuan Wang et al. AgentDropout: Dynamic agent elimination for token-efficient and high-performance LLM-based multi-agent collaboration. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[27] Zhiyuan Weng, Guikun Chen, and Wenguan Wang. Do as we do, not as you think: the conformity of large language models. In *International Conference on Learning Representations (ICLR)*, 2025.

[28] Chenfei Yan et al. When truth is distributed: Misinformation derails collective fact recovery in LLM-based multi-agent systems, 2026. arXiv:2608.03421.

[29] Amotz Zahavi. Mate selection: a selection for a handicap. *Journal of Theoretical Biology*, 53: 205–214, 1975.

[30] Guibin Zhang et al. Cut the crap: An economical communication pipeline for LLM-based multi-agent systems, 2024. arXiv:2410.02506.

[31] Yujie Zhao, Lanxiang Hu, Yang Wang, Minmin Hou, Hao Zhang, Ke Ding, and Jishen Zhao. Stronger-MAS: Multi-agent reinforcement learning for collaborative LLMs, 2025. arXiv:2510.11062.

A Reproducibility

Code, environments and notebooks: https://github.com/Positron00/explRL; a highlights page redraws the main figures from a compact extract of the saved runs (https://positron00.github.io/explRL/highlights.html). Analysis plans were frozen and committed before their runs (fixes: `FIX_PLAN`, hash `79abfb531cdd`; RL: `RL_PLAN` v2, hash `4d1ea0078a6e`; second RL round: `RL13_PLAN`, hash `3e50a4002c7b`; replication of the incentive result: `RL16_PLAN`, hash `082b3b7c9a31`; HiddenBench: `HB_PLAN`, hash `791a2df079a8`, frozen after a pilot, on the benchmark's 65 tasks vendored unchanged under their MIT licence; trust information: `TRUST_PLAN`, hash `6d2818ae84c2`; team size: `SCALE_PLAN`, hash `b886baefb2ef`, frozen after a pilot; traces: `TRACE_PLAN`, hash `f5c25637acd0`; guards: `COLONY_PLAN`, hash `58e842277a3e`; guards at 32 agents: `COLONY32_PLAN`, hash `f51f0298d34a`; guards at 64 agents: `COLONY64_PLAN`, hash `9764cd43e34c`; Claude liars and unassigned roles: `COLONY3_PLAN`, hash `0c59fa10f2b0`; guards that scale: `STAGE4_PLAN`, hash `a441381ce078`; team-size RL: `RL17_PLAN`, hash `8e28b5fdfb1f`; trained roles: `RL18_PLAN`, hash `c103584f5aac`). All API replies are cached locally (available on request), and cached reruns reproduce every row and turn log byte for byte at no cost. Every number in this paper is generated from the saved results by `paper/make_assets.py`, and every figure by `tools/figures.py`.

B A vote fix that did not replicate

In a hidden-profile vote (four agents, shared facts favouring the wrong option, two private facts each favouring the right one, three rounds of posts, then a vote), members posted about half of their private facts (0.50) in every arm, yet a table of the facts posted so far closed 0.75 $[0.50, 1.00]$ of the gap when one member held a decisive fact (win rate 0.88 to 0.97; Holm $p = 0.012$; Figure 5). Rerun unchanged on HiddenBench's 65 tasks [13] (195 episodes per arm), the table did not help (-0.07 $[-0.69, 0.30]$ of the gap); the one arm that moved was an instruction to speak up, which raised disclosure of hidden facts from 0.74 to 0.88 and the win rate from 0.77 to 0.83. Our protocol left little headroom (agent accuracy 0.79 against the benchmark's reported 30%; solo 0.91), and we do not claim the aggregation effect in general.

C A cheap simulation as a screen

Rule-agent simulations in numpy ranked harness designs the way Haiku teams did on a task board (Spearman $\rho = 0.75$ $[0.55, 0.93]$) and on the clue board ($\rho = 0.79$ $[0.74, 0.81]$), but not on text foraging ($\rho = 0.02$ $[-0.47, 0.45]$). Forecasts frozen before the expensive runs were right more often than not but missed the effects that depend on what language models do: guards paying off, larger teams losing more to liars, and guard copies winning at 64 agents (the design-change forecast was right on 4 of its calls, wrong on 1, with 1 no call). We study them as a design-screening method in a companion paper.